Complex Vectors and Matrices

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Complex vectors and matrices

Definition 0.1 (Complex vector)**.** A **complex vector** of length p is a column vector whose entries are complex numbers; the set of all of them is denoted $\mathbb{C}^p$.

Definition 0.2 (Complex matrix)**.** A **complex matrix** of size $m \times n$ is a matrix with complex entries; the set of all of them is denoted $\mathbb{C}^{m \times n}$.

Example 0.1 (A complex vector and its conjugate)**.**

$$\underset{3\times 1}{z} = \begin{bmatrix} 1+i \\ 2 \\ -3\,i \end{bmatrix} \in \mathbb{C}^3, \qquad \underset{3\times 1}{\overline{z}} = \begin{bmatrix} 1-i \\ 2 \\ 3\,i \end{bmatrix}.$$

The real entry 2 is its own conjugate. Every real vector is also a complex vector, since each real number a is the complex number $a + 0\,i$, so $\mathbb{R}^p$ is a subset of $\mathbb{C}^p$.

Definition 0.3 (Complex conjugate of a complex vector or matrix)**.** The **complex conjugate** of a complex vector or matrix takes the complex conjugate of each entry: $(\overline{\mathbf{A}})_{jk} \stackrel{\text{def}}{=} \overline{a_{jk}}$.

Definition 0.4 (Conjugate transpose (Hermitian adjoint, Hermitian conjugate, transjugate))**.** The **conjugate transpose** (also called the **Hermitian adjoint**, **Hermitian conjugate**, or **transjugate**) of an $m \times n$ complex matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^{\mathsf{H}}$ obtained by transposing $\mathbf{A}$ and then taking the complex conjugate of each entry:

$$(\mathbf{A}^{\mathsf{H}})_{jk} \stackrel{\text{def}}{=} \overline{a_{kj}} \quad \text{for } j \in \{1, \dots, n\},\ k \in \{1, \dots, m\}.$$

Remark (Notation for the conjugate transpose)

Remark 0.1 (Notation for the conjugate transpose). The operation is denoted variously across mathematics and physics: $\mathbf{A}^*$ in pure mathematics (Axler (2024, Definition 7.7, p. 231)), $\mathbf{A}^\dagger$ ("A-dagger") in physics and quantum mechanics, and $\mathbf{A}^{\mathsf{H}}$ in engineering, numerical linear algebra, and statistics. These notes write $\mathbf{A}^{\mathsf{H}}$ because * is reserved for optimal points (as in $\underset{\sim}{x}^*$) and † frequently denotes the Moore-Penrose pseudoinverse in the broader literature.

The Hermitian adjoint matrix should not be confused with the classical *adjugate matrix* (the transpose of the matrix of cofactors), which older texts occasionally referred to as the "adjoint."

Example 0.2 (The conjugate transpose of a 2×3 matrix)**.** For

$$\underset{2\times 3}{\mathbf{A}} = \begin{bmatrix} 1+i & 2 & 0 \\ -i & 3-2\,i & 4 \end{bmatrix},$$

transposing gives a 3×2 matrix, and conjugating each of its entries gives

$$\underset{3\times 2}{\mathbf{A}^{\mathsf{H}}} = \begin{bmatrix} 1-i & i \\ 2 & 3+2\,i \\ 0 & 4 \end{bmatrix}.$$

For instance,

$$\begin{aligned} (\mathbf{A}^{\mathsf{H}})_{12} &= \overline{a_{21}} \\ &= \overline{-i} \\ &= i. \end{aligned}$$

The transpose alone would have $-i$ in that position, so for a matrix with an entry that is not real, $\mathbf{A}^{\mathsf{H}}$ and $\mathbf{A}^\top$ differ. For a real matrix, conjugating changes nothing, so $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^\top$.

Remark (Conjugate transpose of a complex vector)

Remark 0.2 (Conjugate transpose of a complex vector). The conjugate transpose of a column vector $\underset{\sim}{z} \in \mathbb{C}^p$ is the $1 \times p$ row vector $\underset{\sim}{z}^{\mathsf{H}} = [\overline{z_1},\ \ldots,\ \overline{z_p}]$.

Theorem 0.1 (Properties of the Hermitian adjoint)**.** *Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be complex matrices of compatible dimensions, and let* $c \in \mathbb{C}$ *be a scalar. The Hermitian adjoint (Definition 0.4) satisfies:*

1. ***Involution:***
$$(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}} = \mathbf{A}$$
2. ***Additivity:***
$$(\mathbf{A} + \mathbf{B})^{\mathsf{H}} = \mathbf{A}^{\mathsf{H}} + \mathbf{B}^{\mathsf{H}}$$
3. ***Conjugate-homogeneity:***
$$(c\,\mathbf{A})^{\mathsf{H}} = \overline{c}\,\mathbf{A}^{\mathsf{H}}$$
4. ***Reversal of products:***
$$(\mathbf{AB})^{\mathsf{H}} = \mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}$$

If $\mathbf{A}$ *is invertible, then* $\mathbf{A}^{\mathsf{H}}$ *is also invertible, and*

$$(\mathbf{A}^{\mathsf{H}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{H}}.$$

Proof

Proof. Properties 1-3 follow directly from entrywise conjugation and transposition:

- For involution:
$$\begin{aligned}[(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}]_{jk} &= \overline{(\mathbf{A}^{\mathsf{H}})_{kj}} \\ &= \overline{\overline{a_{jk}}} \\ &= a_{jk}.\end{aligned}$$
- For additivity:
$$\begin{aligned}[(\mathbf{A}+\mathbf{B})^{\mathsf{H}}]_{jk} &= \overline{(\mathbf{A}+\mathbf{B})_{kj}} \\ &= \overline{a_{kj} + b_{kj}} \\ &= \overline{a_{kj}} + \overline{b_{kj}} \\ &= (\mathbf{A}^{\mathsf{H}})_{jk} + (\mathbf{B}^{\mathsf{H}})_{jk}.\end{aligned}$$
- For conjugate-homogeneity:
$$\begin{aligned}[(c\,\mathbf{A})^{\mathsf{H}}]_{jk} &= \overline{(c\,\mathbf{A})_{kj}} \\ &= \overline{c\,a_{kj}} \\ &= \overline{c}\,\overline{a_{kj}} \\ &= \overline{c}\,(\mathbf{A}^{\mathsf{H}})_{jk}.\end{aligned}$$

  Unlike the real transpose where $(c\,\mathbf{A})^{\top} = c\,\mathbf{A}^{\top}$, scaling conjugates the scalar.

For property 4 (reversal of products), let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ a $k \times n$ matrix. Entry (i, j) of the product's Hermitian adjoint (for $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$) is:

$$\begin{aligned}
[(\mathbf{AB})^{\mathsf{H}}]_{ij} &= \overline{(\mathbf{AB})_{ji}} && \text{(Definition 0.4)} \\
&= \overline{\sum_{s=1}^{k} a_{js}\, b_{si}} && \text{(definition of matrix multiplication)} \\
&= \sum_{s=1}^{k} \overline{a_{js}}\, \overline{b_{si}} && \text{(conjugate of sums and products)} \\
&= \sum_{s=1}^{k} \overline{b_{si}}\, \overline{a_{js}} && \text{(complex multiplication is commutative)} \\
&= \sum_{s=1}^{k} (\mathbf{B}^{\mathsf{H}})_{is}\, (\mathbf{A}^{\mathsf{H}})_{sj} && \text{(Definition 0.4)} \\
&= (\mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}})_{ij} && \text{(definition of matrix multiplication).}
\end{aligned}$$

For the inverse, applying property 4 to $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ gives:

$$\begin{aligned}
(\mathbf{A}^{-1})^{\mathsf{H}}\mathbf{A}^{\mathsf{H}} &= (\mathbf{A}\mathbf{A}^{-1})^{\mathsf{H}} \\
&= \mathbf{I}^{\mathsf{H}} \\
&= \mathbf{I},
\end{aligned}$$

so $(\mathbf{A}^{\mathsf{H}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{H}}$. □

Theorem 0.2 (The adjoint property for inner products)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *complex matrix. Under the standard inner product on complex Euclidean space,* $\left\langle \underset{\sim}{u}, \underset{\sim}{v} \right\rangle \overset{def}{=} \underset{\sim}{u}^{\mathsf{H}} \underset{\sim}{v}$*,*

$$\left\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v} \right\rangle = \left\langle \underset{\sim}{u}, \mathbf{A}^{\mathsf{H}}\underset{\sim}{v} \right\rangle \quad \textit{for all } \underset{\sim}{u} \in \mathbb{C}^n \textit{ and } \underset{\sim}{v} \in \mathbb{C}^m.$$

Proof

Proof. By the definition of the standard complex inner product and the product rule for the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}
\left\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v} \right\rangle &= (\mathbf{A}\underset{\sim}{u})^{\mathsf{H}}\underset{\sim}{v} && \text{(definition of standard inner product)} \\
&= (\underset{\sim}{u}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}})\underset{\sim}{v} && \text{(Theorem 0.1, reversal of products)} \\
&= \underset{\sim}{u}^{\mathsf{H}}(\mathbf{A}^{\mathsf{H}}\underset{\sim}{v}) && \text{(associativity of matrix multiplication)} \\
&= \left\langle \underset{\sim}{u}, \mathbf{A}^{\mathsf{H}}\underset{\sim}{v} \right\rangle && \text{(definition of standard inner product).}
\end{aligned}$$

□

Remark (Where the name "adjoint" comes from)

Remark 0.3 (Where the name "adjoint" comes from). This identity is the origin of the name **adjoint**: in functional analysis and operator theory, the *adjoint* of a linear map $T : V \to W$ between inner product spaces is defined as the unique operator $T^* : W \to V$ satisfying $\left\langle T\underset{\sim}{u}, \underset{\sim}{v} \right\rangle = \left\langle \underset{\sim}{u}, T^*\underset{\sim}{v} \right\rangle$ (Axler (2024, Definition 7.1, p. 226)). When V and W are finite-dimensional spaces equipped with their standard orthonormal bases, the matrix representing the adjoint operator T^* is precisely the Hermitian adjoint matrix $\mathbf{A}^{\mathsf{H}}$ (Axler (2024, Proposition 7.9, p. 231)).

Theorem 0.3 ($\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z}$ is a sum of squared absolute values)**.** *For every* $\underset{\sim}{z} \in \mathbb{C}^p$,

$$\underset{1\times p}{\underset{\sim}{z}^{\mathsf{H}}}\ \underset{p\times 1}{\underset{\sim}{z}} = \sum_{j=1}^{p} |z_j|^2,$$

which is a nonnegative real number, and is 0 *only when* $\underset{\sim}{z} = \underset{\sim_{p\times 1}}{0}$.

Proof

Proof. The last step uses the identity for a complex number times its conjugate.

$$\begin{aligned}
\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z} &= \sum_{j=1}^{p} \overline{z_j}\, z_j && \text{(Definition 0.4, and the row-times-column product)} \\
&= \sum_{j=1}^{p} z_j\, \overline{z_j} && \text{(complex multiplication is commutative)} \\
&= \sum_{j=1}^{p} |z_j|^2 && \text{(a number times its conjugate)}
\end{aligned}$$

By the definition of the absolute value, each $|z_j|^2 = (\operatorname{Re} z_j)^2 + (\operatorname{Im} z_j)^2$ is a nonnegative real number, so their sum is too. A sum of nonnegative numbers is 0 only when every term is 0, and $|z_j|^2 = 0$ only when

$$\begin{aligned}
\operatorname{Re} z_j &= \operatorname{Im} z_j \\
&= 0,
\end{aligned}$$

that is, when $z_j = 0$. □

Example 0.3 (Computing $\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z}$)**.** Let $\underset{\sim}{z} = (1+i, 2)^{\top}$. By Theorem 0.3,

$$\begin{aligned}
\underbrace{\underset{\sim}{z}^{\mathsf{H}}}_{1\times 2}\underbrace{\underset{\sim}{z}}_{2\times 1} &= |1+i|^2 + |2|^2 && \text{(Theorem 0.3)} \\
&= (1^2+1^2) + (2^2+0^2) && \text{(definition of the absolute value, squared)} \\
&= 2+4 && \text{(add inside each group)} \\
&= 6 && \text{(add)}
\end{aligned}$$

Multiplying out directly gives the same number: $\overline{2}\cdot 2 = 4$, and

$$\begin{aligned}
\overline{(1+i)}\,(1+i) &= (1-i)(1+i) && \text{(conjugate)} \\
&= 1+i-i-i^2 && \text{(distribute)} \\
&= 1-i^2 && \text{(cancel } i\text{)} \\
&= 2 && (i^2=-1)
\end{aligned}$$

Example 0.4 (Why complex vectors use the conjugate transpose)**.** For a real vector $\underset{\sim}{x}$, $\underset{\sim}{x}^{\top}\underset{\sim}{x} = \sum_{j=1}^{p} x_j^2$ is 0 only when $\underset{\sim}{x}$ is the zero vector. The same formula fails for complex vectors. Let $\underset{\sim}{z} = (1, i)^{\top}$. Then

$$\begin{aligned}
\underbrace{\underset{\sim}{z}^{\top}}_{1\times 2}\underbrace{\underset{\sim}{z}}_{2\times 1} &= 1\cdot 1 + i\cdot i && \text{(multiply entry by entry and add)} \\
&= 1+i^2 && \text{(multiply)} \\
&= 0 && (i^2=-1)
\end{aligned}$$

although $\underset{\sim}{z} \neq \underset{\sim 2\times 1}{0}$. Using the conjugate transpose instead,

$$\begin{aligned}
\underbrace{\underset{\sim}{z}^{\mathsf{H}}}_{1\times 2}\underbrace{\underset{\sim}{z}}_{2\times 1} &= \overline{1}\cdot 1 + \overline{i}\cdot i && \text{(Definition 0.4)} \\
&= 1+(-i)\cdot i && \text{(conjugate each entry)} \\
&= 1-i^2 && \text{(multiply)} \\
&= 2 && (i^2=-1)
\end{aligned}$$

which is positive, as Theorem 0.3 guarantees.

Definition 0.5 (Hermitian matrix (self-adjoint matrix))**.** A square complex matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is **Hermitian** (or **self-adjoint**) if it equals its conjugate transpose (Definition 0.4):

$$\mathbf{A}^{\mathsf{H}} = \mathbf{A}.$$

Remark (Why "self-adjoint")

Remark 0.4 (Why "self-adjoint"). The term *self-adjoint* comes directly from Theorem 0.2: when $\mathbf{A}$ is Hermitian, $\left\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v} \right\rangle = \left\langle \underset{\sim}{u}, \mathbf{A}\underset{\sim}{v} \right\rangle$ for all $\underset{\sim}{u}, \underset{\sim}{v} \in \mathbb{C}^n$. The linear operator moves across the inner product without change.

Example 0.5 (A Hermitian matrix)**.**

$$\underset{2\times 2}{\mathbf{A}} = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$$

is Hermitian: transposing swaps the off-diagonal entries, and conjugating turns $1+i$ back into $1-i$ and $1-i$ back into $1+i$, so $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$. The diagonal entries of a Hermitian matrix must be real, because $a_{jj} = \overline{a_{jj}}$, and only a real number is its own conjugate: if $a + b\,i = a - b\,i$, then the imaginary parts give $b = -b$, so $b = 0$.
Every real symmetric matrix is Hermitian, because for a real matrix $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\top}$.

Example 0.6 (A symmetric complex matrix that is not Hermitian)**.**

$$\underset{2\times 2}{\mathbf{B}} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

is symmetric, $\mathbf{B}^{\top} = \mathbf{B}$, but it is not Hermitian:

$$\underset{2\times 2}{\mathbf{B}^{\mathsf{H}}} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \neq \mathbf{B}.$$

Definition 0.6 (Skew-Hermitian matrix (anti-Hermitian matrix))**.** A square complex matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is **skew-Hermitian** (or **anti-Hermitian**) if it equals the negative of its conjugate transpose (Definition 0.4):

$$\mathbf{A}^{\mathsf{H}} = -\mathbf{A}.$$

Remark (The diagonal of a skew-Hermitian matrix)

Remark 0.5 (The diagonal of a skew-Hermitian matrix). Every entry of a skew-Hermitian matrix satisfies $a_{jk} = -\overline{a_{kj}}$. In particular, its diagonal entries satisfy $a_{jj} = -\overline{a_{jj}}$, which means

$$\begin{aligned} a_{jj} + \overline{a_{jj}} &= 2\,\mathrm{Re}(a_{jj}) \\ &= 0. \end{aligned}$$

Therefore, every diagonal entry of a skew-Hermitian matrix must be purely imaginary (or zero). For example, $a_{jj} = 3\,i$ works, since

$$\begin{aligned} -\overline{3\,i} &= -(-3\,i) \\ &= 3\,i, \end{aligned}$$

but $a_{jj} = 3$ does not, since $-\overline{3} = -3$.

Example 0.7 (A skew-Hermitian matrix)**.** The matrix

$$\mathbf{A} = \begin{bmatrix} 2\,i & 1+i \\ -1+i & 0 \end{bmatrix}$$

is skew-Hermitian: transposing and conjugating gives

$$\begin{aligned} \mathbf{A}^{\mathsf{H}} &= \begin{bmatrix} \overline{2\,i} & \overline{-1+i} \\ \overline{1+i} & \overline{0} \end{bmatrix} \\ &= \begin{bmatrix} -2\,i & -1-i \\ 1-i & 0 \end{bmatrix} \\ &= -\mathbf{A}. \end{aligned}$$

Its diagonal entries $2\,i$ and 0 are purely imaginary.

Theorem 0.4 (Hermitian and skew-Hermitian decomposition)**.** *Every square complex matrix* $\mathbf{A} \in \mathbb{C}^{n\times n}$ *can be uniquely expressed as the sum of a Hermitian matrix* $\mathbf{H}$ *(Definition 0.5) and a skew-Hermitian matrix* $\mathbf{S}$ *(Definition 0.6):*

$$\mathbf{A} = \mathbf{H} + \mathbf{S},$$

where

$$\mathbf{H} = \frac{\mathbf{A} + \mathbf{A}^{\mathsf{H}}}{2} \quad \textit{and} \quad \mathbf{S} = \frac{\mathbf{A} - \mathbf{A}^{\mathsf{H}}}{2}.$$

Proof

Proof. **Existence**: Define $\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{H}})$ and $\mathbf{S} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^{\mathsf{H}})$. Their sum is:

$$\begin{aligned}\mathbf{H}+\mathbf{S} &= \frac{\mathbf{A}+\mathbf{A}^{\mathsf{H}}}{2}+\frac{\mathbf{A}-\mathbf{A}^{\mathsf{H}}}{2}\\ &= \frac{2\mathbf{A}}{2}\\ &= \mathbf{A}.\end{aligned}$$

Applying the properties of the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}\mathbf{H}^{\mathsf{H}} &= \left(\frac{\mathbf{A}+\mathbf{A}^{\mathsf{H}}}{2}\right)^{\mathsf{H}}\\ &= \frac{\mathbf{A}^{\mathsf{H}}+(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}}{2}\\ &= \frac{\mathbf{A}^{\mathsf{H}}+\mathbf{A}}{2}\\ &= \mathbf{H},\\ \mathbf{S}^{\mathsf{H}} &= \left(\frac{\mathbf{A}-\mathbf{A}^{\mathsf{H}}}{2}\right)^{\mathsf{H}}\\ &= \frac{\mathbf{A}^{\mathsf{H}}-(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}}{2}\\ &= \frac{\mathbf{A}^{\mathsf{H}}-\mathbf{A}}{2}\\ &= -\mathbf{S}.\end{aligned}$$

Thus $\mathbf{H}$ is Hermitian and $\mathbf{S}$ is skew-Hermitian.
Uniqueness: Suppose $\mathbf{A}=\mathbf{H}'+\mathbf{S}'$ with $(\mathbf{H}')^{\mathsf{H}}=\mathbf{H}'$ and $(\mathbf{S}')^{\mathsf{H}}=-\mathbf{S}'$. Taking the Hermitian adjoint of both sides yields:

$$\begin{aligned}\mathbf{A}^{\mathsf{H}} &= (\mathbf{H}'+\mathbf{S}')^{\mathsf{H}}\\ &= (\mathbf{H}')^{\mathsf{H}}+(\mathbf{S}')^{\mathsf{H}}\\ &= \mathbf{H}'-\mathbf{S}'.\end{aligned}$$

Adding $\mathbf{A}=\mathbf{H}'+\mathbf{S}'$ and $\mathbf{A}^{\mathsf{H}}=\mathbf{H}'-\mathbf{S}'$ gives $\mathbf{A}+\mathbf{A}^{\mathsf{H}}=2\mathbf{H}'$, so

$$\begin{aligned}\mathbf{H}' &= \frac{1}{2}(\mathbf{A}+\mathbf{A}^{\mathsf{H}})\\ &= \mathbf{H}.\end{aligned}$$

Subtracting the two equations gives $\mathbf{A}-\mathbf{A}^{\mathsf{H}}=2\mathbf{S}'$, so

$$\begin{aligned}\mathbf{S}' &= \frac{1}{2}(\mathbf{A}-\mathbf{A}^{\mathsf{H}})\\ &= \mathbf{S}.\end{aligned}$$

□

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